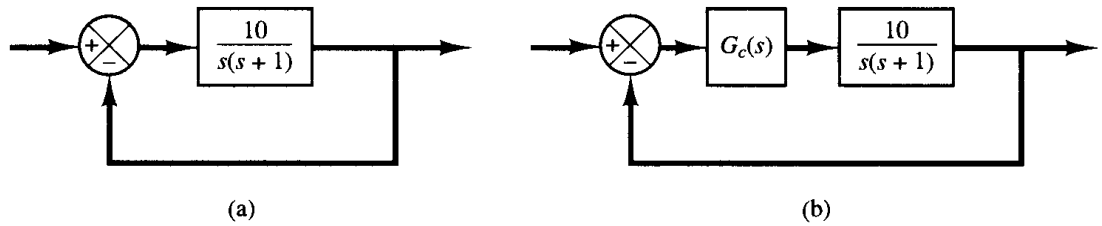


**Figure 9–43**

(a) Control system;  
(b) compensated system.



Define

$$G_1(s) = KG(s) = \frac{10K}{s(s+1)}$$

where  $K = K_c\alpha$ .

The first step in the design is to adjust the gain  $K$  to meet the steady-state performance specification or to provide the required static velocity error constant. Since the static velocity error constant  $K_v$  is given as  $20 \text{ sec}^{-1}$ , we have

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG_c(s)G(s) \\ &= \lim_{s \rightarrow 0} s \frac{Ts+1}{aTs+1} G_1(s) \\ &= \lim_{s \rightarrow 0} \frac{s10K}{s(s+1)} \\ &= 10K = 20 \end{aligned}$$

or

$$K = 2$$

With  $K = 2$ , the compensated system will satisfy the steady-state requirement.

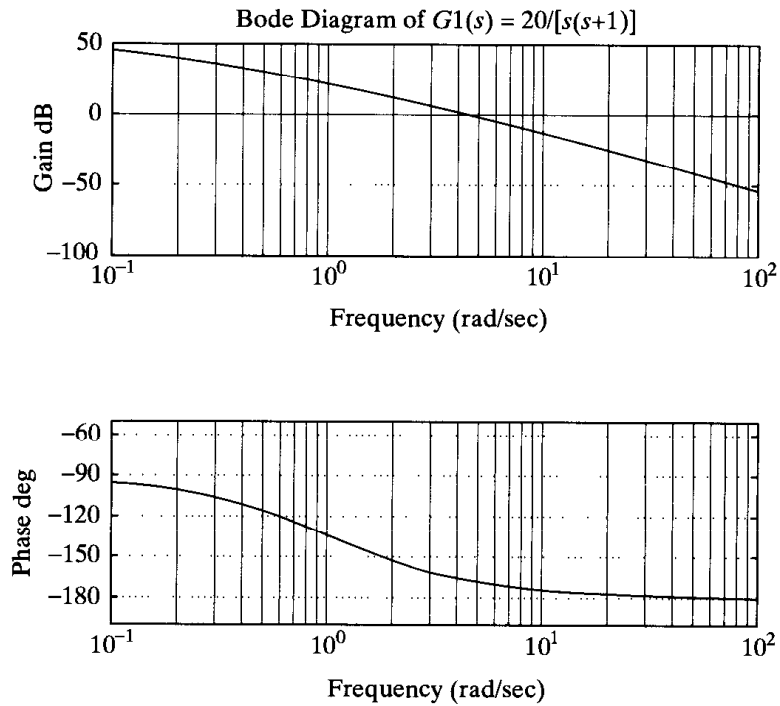
We shall next plot the Bode diagram of

$$G_1(s) = \frac{20}{s(s+1)}$$

MATLAB Program 9–6 produces the Bode diagram shown in Figure 9–44. From this plot, the phase margin is found to be  $14^\circ$ . The gain margin is  $+\infty \text{ dB}$ .

**MATLAB Program 9–6**

```
num = [0 0 20];  
den = [1 1 0];  
w = logspace(-1,2,100);  
bode(num,den,w)  
subplot(2,1,1);  
title('Bode Diagram of G1(s) = 20/[s(s + 1)]')
```



**Figure 9-44**  
Bode diagram of  $G_1(s)$ .

Since the specification calls for a phase margin of  $50^\circ$ , the additional phase lead necessary to satisfy the phase-margin requirement is  $36^\circ$ . A lead compensator can contribute this amount.

Noting that the addition of a lead compensator modifies the magnitude curve in the Bode diagram, we realize that the gain crossover frequency will be shifted to the right. We must offset the increased phase lag of  $G_1(j\omega)$  due to this increase in the gain crossover frequency. Taking the shift of the gain crossover frequency into consideration, we may assume that  $\phi_m$ , the maximum phase lead required, is approximately  $41^\circ$ . (This means that approximately  $5^\circ$  has been added to compensate for the shift in the gain crossover frequency.) Since

$$\sin \phi_m = \frac{1 - \alpha}{1 + \alpha}$$

$\phi_m = 41^\circ$  corresponds to  $\alpha = 0.2077$ . Note that  $\alpha = 0.21$  corresponds to  $\phi_m = 40.76^\circ$ . Whether we choose  $\phi_m = 41^\circ$  or  $\phi_m = 40.76^\circ$  does not make much difference in the final solution. Hence, let us choose  $\alpha = 0.21$ .

Once the attenuation factor  $\alpha$  has been determined on the basis of the required phase-lead angle, the next step is to determine the corner frequencies  $\omega = 1/T$  and  $\omega = 1/(\alpha T)$  of the lead compensator. Notice that the maximum phase-lead angle  $\phi_m$  occurs at the geometric mean of the two corner frequencies, or  $\omega = 1/(\sqrt{\alpha}T)$ .

The amount of the modification in the magnitude curve at  $\omega = 1/(\sqrt{\alpha}T)$  due to the inclusion of the term  $(Ts + 1)/(\alpha Ts + 1)$  is

$$\left| \frac{1 + j\omega T}{1 + j\omega \alpha T} \right|_{\omega = \frac{1}{\sqrt{\alpha}T}} = \left| \frac{1 + j \frac{1}{\sqrt{\alpha}}}{1 + j \alpha \frac{1}{\sqrt{\alpha}}} \right| = \frac{1}{\sqrt{\alpha}}$$

Note that

$$\frac{1}{\sqrt{\alpha}} = \frac{1}{\sqrt{0.21}} = 6.7778 \text{ dB}$$

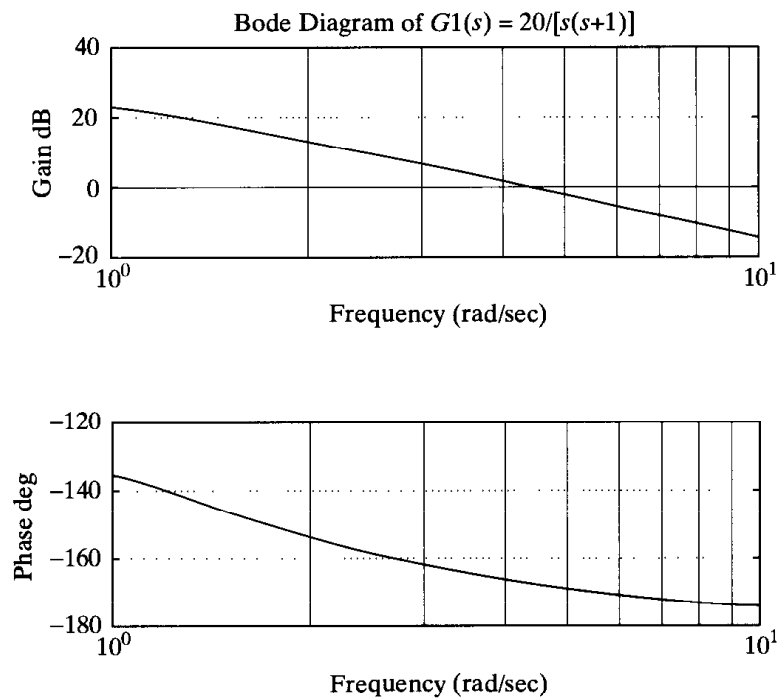
We need to find the frequency point where, when the lead compensator is added, the total magnitude becomes 0 dB.

From Figure 9–44 we see that the frequency point where the magnitude of  $G_1(j\omega)$  is  $-6.7778$  dB occurs between  $\omega = 1$  and  $10$  rad/sec. Hence, we plot a new Bode diagram of  $G_1(j\omega)$  in the frequency range between  $\omega = 1$  and  $10$  to locate the exact point where  $G_1(j\omega) = -6.7778$  dB. MATLAB Program 9–7 produces the Bode diagram in this frequency range, which is shown in

|                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATLAB Program 9–7                                                                                                                                    |
| <pre> num = [0  0  20]; den = [1  1  0]; w = logspace(0,1,100); bode(num,den,w) subplot(2,1,1); title('Bode Diagram of G1(s) = 20/[s(s + 1)]') </pre> |

Figure 9–45. From this diagram, we find the frequency point where  $|G_1(j\omega)| = -6.7778$  dB occurs at  $\omega = 6.5$  rad/sec. Let us select this frequency to be the new gain crossover frequency, or  $\omega_c = 6.5$  rad/sec. Noting that this frequency corresponds to  $1/(\sqrt{\alpha}T)$ , or

$$\omega_c = \frac{1}{\sqrt{\alpha}T}$$



**Figure 9–45**  
Bode diagram of  
 $G_1(s)$ .

we obtain

$$\frac{1}{T} = \omega_c \sqrt{\alpha} = 6.5 \sqrt{0.21} = 2.9787$$

and

$$\frac{1}{\alpha T} = \frac{\omega_c}{\sqrt{\alpha}} = \frac{6.5}{\sqrt{0.21}} = 14.1842$$

The lead compensator thus determined is

$$G_c(s) = K_c \frac{s + 2.9787}{s + 14.1842} = K_c \alpha \frac{0.3357s + 1}{0.07050s + 1}$$

where  $K_c$  is determined as

$$K_c = \frac{K}{\alpha} = \frac{2}{0.21} = 9.5238$$

Thus, the transfer function of the compensator becomes

$$G_c(s) = 9.5238 \frac{s + 2.9787}{s + 14.1842} = 2 \frac{0.3357s + 1}{0.07050s + 1}$$

MATLAB Program 9–8 produces the Bode diagram of this lead compensator, which is shown in Figure 9–46.

MATLAB Program 9–8

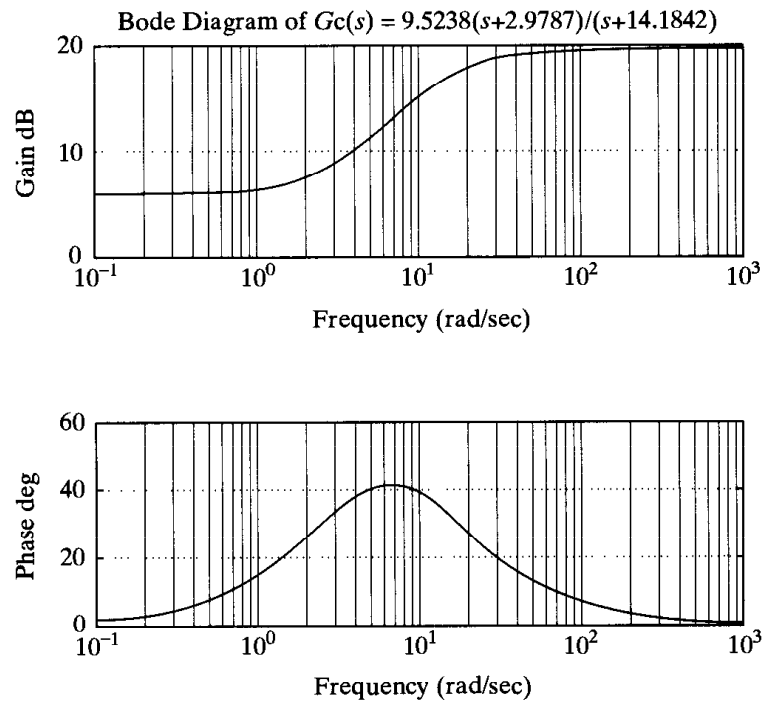
```
numc = [9.5238 28.3685];
denc = [1 14.1842];
w = logspace(-1,3,100);
bode(numc,denc,w)
subplot(2,1,1);
title('Bode Diagram of Gc(s) = 9.5238(s + 2.9787)/(s + 14.1842)')
```

The open-loop transfer function of the designed system is

$$\begin{aligned} G_c(s)G(s) &= 9.5238 \frac{s + 2.9787}{s + 14.1842} \frac{10}{s(s + 1)} \\ &= \frac{95.238s + 283.6854}{s^3 + 15.1842s^2 + 14.1842s} \end{aligned}$$

MATLAB Program 9–9 will produce the Bode diagram of  $G_c(s)G(s)$ , which is shown in Figure 9–47.

From Figure 9–47 it is clearly seen that the phase margin is approximately  $50^\circ$  and the gain margin is  $+\infty$  dB. Since the static velocity error constant  $K_v$  is  $20 \text{ sec}^{-1}$ , all the specifications are met. Before we conclude this problem, we need to check the transient-response characteristics.



**Figure 9–46**  
Bode diagram of  
 $G_c(s)$ .

Unit-step response: We shall compare the unit-step response of the compensated system and that of the original uncompensated system.

MATLAB Program 9–9

```
num = [0 0 95.238 283.6854];
den = [1 15.1842 14.1842 0];
w = logspace(-1,3,100);
bode(num,den,w)
subplot(2,1,1);
title('Bode Diagram of  $G_c(s)G(s)$ ')
```

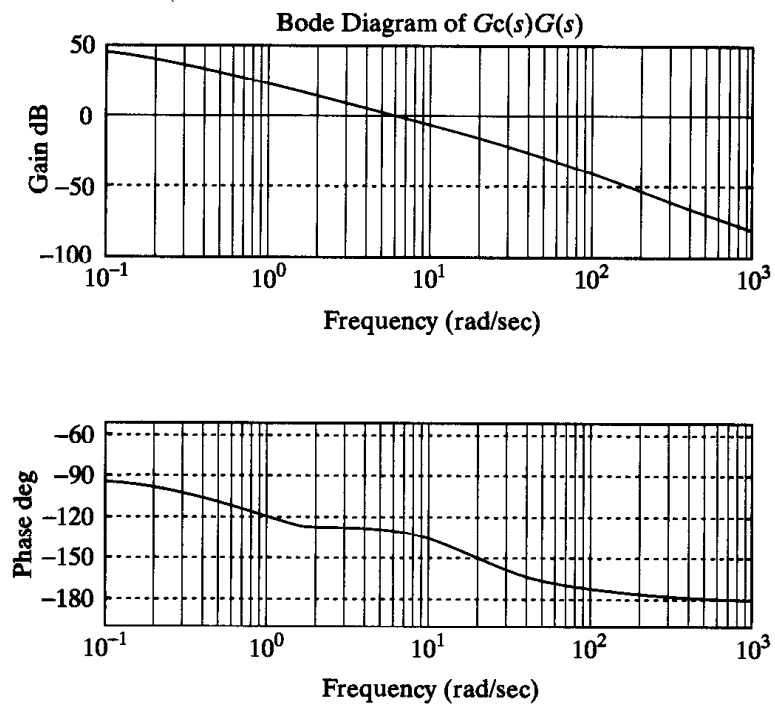
The closed-loop transfer function of the original uncompensated system is

$$\frac{C(s)}{R(s)} = \frac{10}{s^2 + s + 10}$$

The closed-loop transfer function of the compensated system is

$$\frac{C(s)}{R(s)} = \frac{95.238s + 283.6854}{s^3 + 15.1842s^2 + 109.4222s + 283.6854}$$

MATLAB Program 9–10 produces the unit-step responses of the uncompensated and compensated systems. The resulting response curves are shown in Figure 9–48. Clearly, the compensated system exhibits a satisfactory response. Note that the closed-loop zero and poles are located as follows:



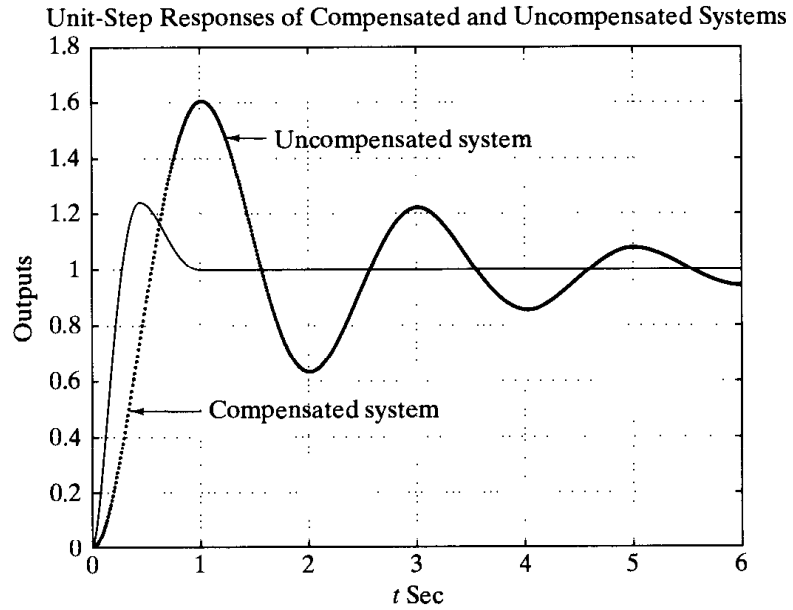
**Figure 9–47**  
Bode diagram of  
 $G_c(s)G(s)$ .

**MATLAB Program 9–10**

```
%*****Unit-step responses*****

num1 = [0  0  10];
den1 = [1  1  10];
num2 = [0  0  95.238  283.6854];
den2 = [1  15.1842  109.4222  283.6854];
t = 0:0.01:6;
[c1,x1,t] = step(num1,den1,t);
[c2,x2,t] = step(num2,den2,t);
plot(t,c1,'.',t,c2,'-')
grid
title('Unit-Step Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs')
text(1.1,0.5,'Compensated system')
text(1.7,1.46,'Uncompensated system')
```

**Figure 9–48**  
Unit-step responses  
of the compensated  
and uncompensated  
systems.



Zero at  $s = -2.9787$

Poles at  $s = -5.2270 \pm j5.7141$ ,  $s = -4.7303$

**Unit-ramp response:** It is worthwhile to check the unit-ramp response of the compensated system. Since  $K_v = 20 \text{ sec}^{-1}$ , the steady-state error following the unit-ramp input will be  $1/K_v = 0.05$ . The static velocity error constant of the uncompensated system is  $10 \text{ sec}^{-1}$ . Hence, the original uncompensated system will have twice as large a steady-state error in following the unit-ramp input.

MATLAB Program 9–11 produces the unit-ramp response curves. [Note that the unit-ramp response is obtained as the unit-step response of  $C(s)/sR(s)$ .] The resulting curves are shown in Figure 9–49. The compensated system has a steady-state error equal to one-half that of the original uncompensated system.

**A-9-9.** Consider the unity feedback system whose open-loop transfer function is

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

Design a compensator  $G_c(s)$  such that the static velocity error constant is  $10 \text{ sec}^{-1}$ , the phase margin is  $50^\circ$ , and the gain margin is 10 dB or more.

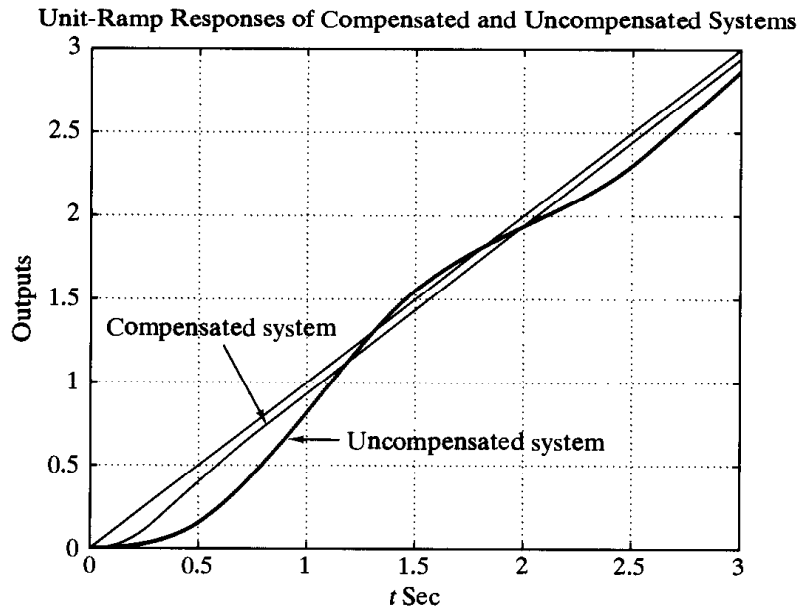
**Solution.** We shall design a lag-lead compensator of the form

$$G_c(s) = K_c \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)}$$

# MATLAB Program 9-11

```
%*****Unit-ramp responses*****

num1 = [0 0 0 10];
den1 = [1 1 10 0];
num2 = [0 0 0 95.238 283.6854];
den2 = [1 15.1842 109.4222 283.6854 0];
t = 0:0.01:3;
[c1,x1,t] = step(num1,den1,t);
[c2,x2,t] = step(num2,den2,t);
plot(t,c1,'-',t,c2,'-',t,t,'- -')
grid
title('Unit-Ramp Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs')
text(0.07,1.3,'Compensated system')
text(1.2,0.65,'Uncompensated system')
```



**Figure 9-49**  
Unit-ramp responses  
of the compensated  
and uncompensated  
systems.

Then the open-loop transfer function of the compensated system is  $G_c(s)G(s)$ . Since the gain  $K$  of the plant is adjustable, let us assume that  $K_c = 1$ . Then  $\lim_{s \rightarrow 0} G_c(s) = 1$ . From the requirement on the static velocity error constant, we obtain

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG_c(s)G(s) = \lim_{s \rightarrow 0} sG_c(s) \frac{K}{s(s+1)(s+4)} \\ &= \frac{K}{4} = 10 \end{aligned}$$



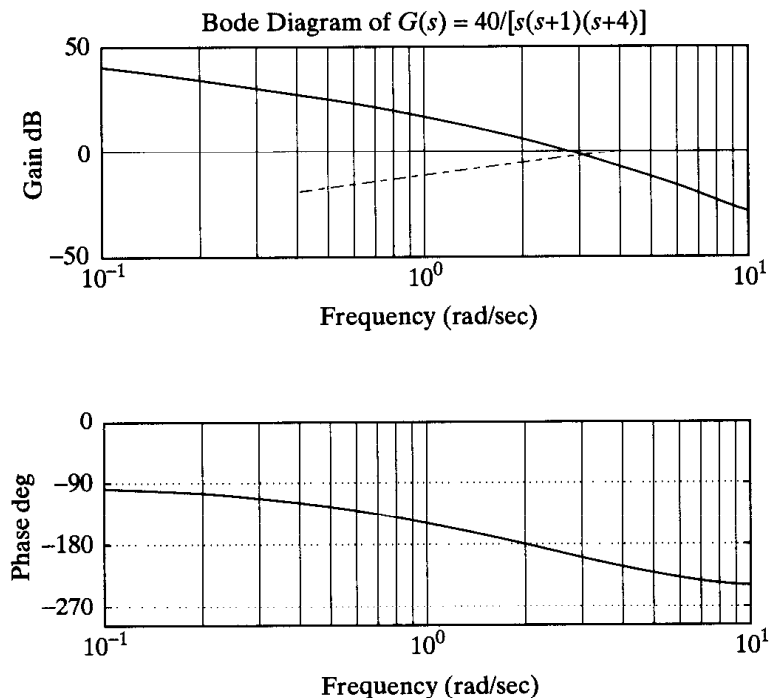
Hence,

$$K = 40$$

We shall first plot a Bode diagram of the uncompensated system with  $K = 40$ . MATLAB Program 9-12 may be used to plot this Bode diagram. The diagram obtained is shown in Figure 9-50.

| MATLAB Program 9-12                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> num = [0  0  0  40]; den = [1  5  4  0]; w = logspace(-1,1,100); bode(num,den,w) subpole(2,1,1); title('Bode Diagram of G(s) = 40/[s(s + 1)(s + 4)]') </pre> |

From Figure 9-50, the phase margin of the uncompensated system is found to be  $-16^\circ$ , which indicates that the uncompensated system is unstable. The next step in the design of a lag-lead compensator is to choose a new gain crossover frequency. From the phase-angle curve for  $G(j\omega)$ , we notice that the phase crossover frequency is  $\omega = 2$  rad/sec. We may choose the new gain crossover frequency to be 2 rad/sec so that the phase-lead angle required at  $\omega = 2$  rad/sec is about  $50^\circ$ . A single lag-lead compensator can provide this amount of phase-lead angle quite easily.



**Figure 9-50**  
Bode diagram  
of  $G(s) =$   
 $40/[s(s + 1)(s + 4)]$ .

Once we choose the gain crossover frequency to be 2 rad/sec, we can determine the corner frequencies of the phase-lag portion of the lag-lead compensator. Let us choose the corner frequency  $\omega = 1/T_2$  (which corresponds to the zero of the phase-lag portion of the compensator) to be 1 decade below the new gain crossover frequency, or at  $\omega = 0.2$  rad/sec. For another corner frequency  $\omega = 1/(\beta T_2)$ , we need the value of  $\beta$ . The value of  $\beta$  can be determined from the consideration of the lead portion of the compensator, as shown next.

For the lead compensator, the maximum phase-lead angle  $\phi_m$  is given by

$$\sin \phi_m = \frac{\beta - 1}{\beta + 1}$$

Notice that  $\beta = 10$  corresponds to  $\phi_m = 54.9^\circ$ . Since we need a  $50^\circ$  phase margin, we may choose  $\beta = 10$ . (Note that we will be using several degrees less than the maximum angle,  $54.9^\circ$ .) Thus,

$$\beta = 10$$

Then the corner frequency  $\omega = 1/(\beta T_2)$  (which corresponds to the pole of the phase-lag portion of the compensator) becomes

$$\omega = 0.02$$

The transfer function of the phase-lag portion of the lag-lead compensator becomes

$$\frac{s + 0.2}{s + 0.02} = 10 \left( \frac{5s + 1}{50s + 1} \right)$$

The phase-lead portion can be determined as follows: Since the new gain crossover frequency is  $\omega = 2$  rad/sec, from Figure 9-50,  $|G(j2)|$  is found to be 6 dB. Hence, if the lag-lead compensator contributes  $-6$  dB at  $\omega = 1$  rad/sec, then the new gain crossover frequency is as desired. From this requirement, it is possible to draw a straight line of slope 20 dB/decade passing through the point  $(-6 \text{ dB}, 2 \text{ rad/sec})$ . (Such a line has been manually drawn on Figure 9-50.) The intersections of this line and the 0-dB line and  $-20$ -dB line determine the corner frequencies. From this consideration, the corner frequencies for the lead portion can be determined as  $\omega = 0.4$  rad/sec and  $\omega = 4$  rad/sec. Thus, the transfer function of the lead portion of the lag-lead compensator becomes

$$\frac{s + 0.4}{s + 4} = \frac{1}{10} \left( \frac{2.5s + 1}{0.25s + 1} \right)$$

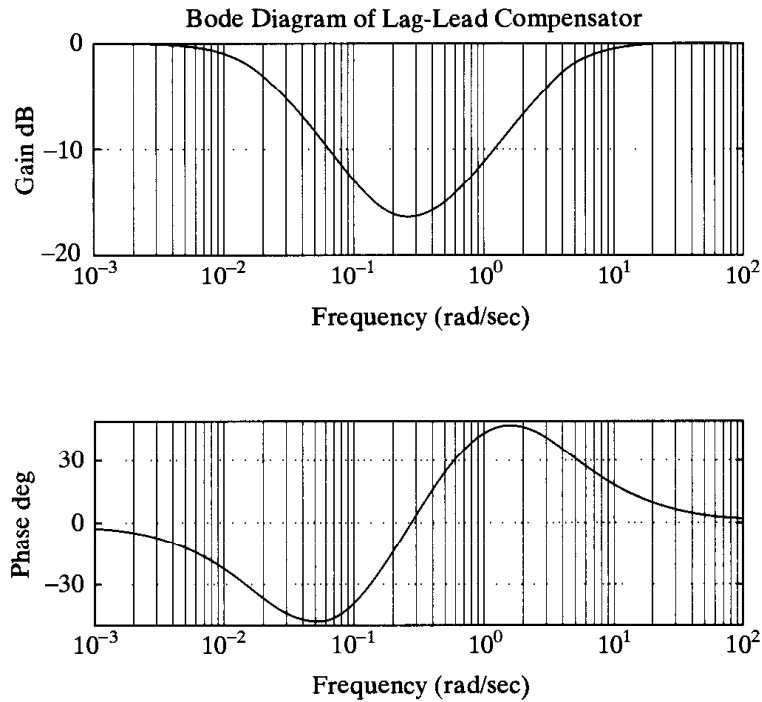
Combining the transfer functions of the lag and lead portions of the compensator, we can obtain the transfer function  $G_c(s)$  of the lag-lead compensator. Since we chose  $K_c = 1$ , we have

$$G_c(s) = \frac{s + 0.4}{s + 4} \frac{s + 0.2}{s + 0.02} = \frac{(2.5s + 1)(5s + 1)}{(0.25s + 1)(50s + 1)}$$

The Bode diagram of the lag-lead compensator  $G_c(s)$  can be obtained by entering MATLAB Program 9-13 into the computer. The resulting plot is shown in Figure 9-51.

#### MATLAB Program 9-13

```
numc = [1  0.6  0.08];
denc = [1  4.02  0.08];
bode(numc,denc)
subplot(2,1,1);
title('Bode Diagram of Lag-Lead Compensator')
```



**Figure 9-51**  
Bode diagram of the  
designed lag-lead  
compensator.

The open-loop transfer function of the compensated system is

$$G_c(s)G(s) = \frac{(s + 0.4)(s + 0.2)}{(s + 4)(s + 0.02)} \frac{40}{s(s + 1)(s + 4)}$$

$$= \frac{40s^2 + 24s + 3.2}{s^5 + 9.02s^4 + 24.18s^3 + 16.48s^2 + 0.32s}$$

Using MATLAB Program 9-14 the magnitude and phase-angle curves of the designed open-loop transfer function  $G_c(s)G(s)$  can be obtained as shown in Figure 9-52. Note that the denominator polynomial den was obtained using the conv command, as follows:

```
a = [1  4.02  0.08];
b = [1  5  4  0];
conv(a,b)

ans =

    1.0000    9.0200   24.1800   16.4800   0.320000    0
```

From Figure 9-52, we see that the actual gain crossover frequency is slightly shifted from 2 rad/sec to a lower value. The actual gain crossover frequency can be found by plotting the Bode diagram in the region  $1 \leq \omega \leq 10$ . It is found to be  $\omega = 1.86$  rad/sec. [Such a small shift of the gain crossover frequency from the assumed gain crossover frequency (2 rad/sec in this case) always occurs in the present method of design.]

# MATLAB Program 9-14

```
num1 = [0 0 0 40 24 3.2];
den1 = [1 9.02 24.18 16.48 0.32 0];
bode(num1,den1)
subplot(2,1,1);
title('Bode Diagram of Gc(s)G(s)')
```

Since the phase margin of the compensated system is  $50^\circ$ , the gain margin is 12.5 dB, and the static velocity error constant is  $10 \text{ sec}^{-1}$ , all the requirements are met.

Figure 9-53 shows the Nyquist plots of  $G(j\omega)$  (uncompensated case) and  $G_c(j\omega)G(j\omega)$  (compensated case). MATLAB Program 9-15 was used to obtain Figure 9-53.

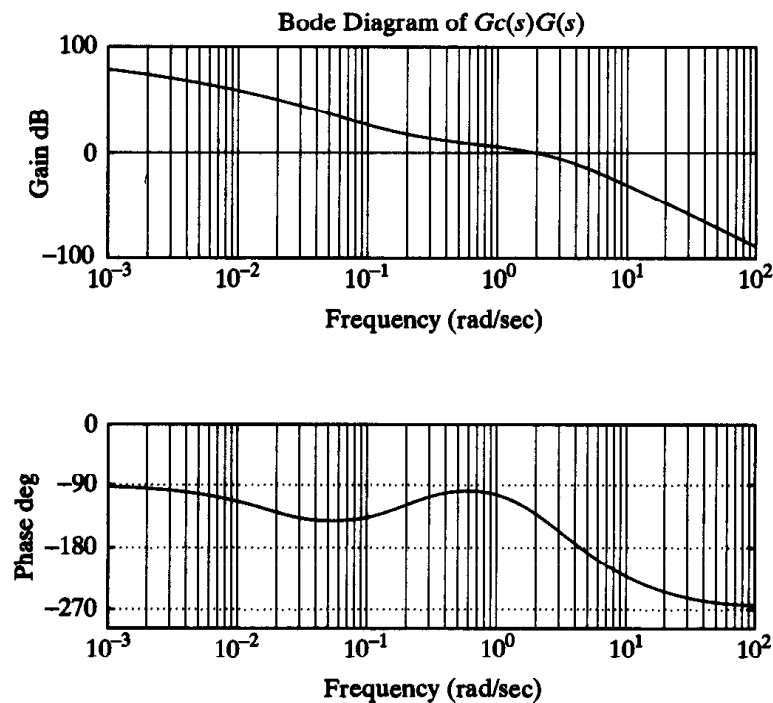
We shall next investigate the transient-response characteristics of the designed system.

Unit-step response: Noting that

$$G_c(s)G(s) = \frac{40(s + 0.4)(s + 0.2)}{(s + 4)(s + 0.02)s(s + 1)(s + 4)}$$

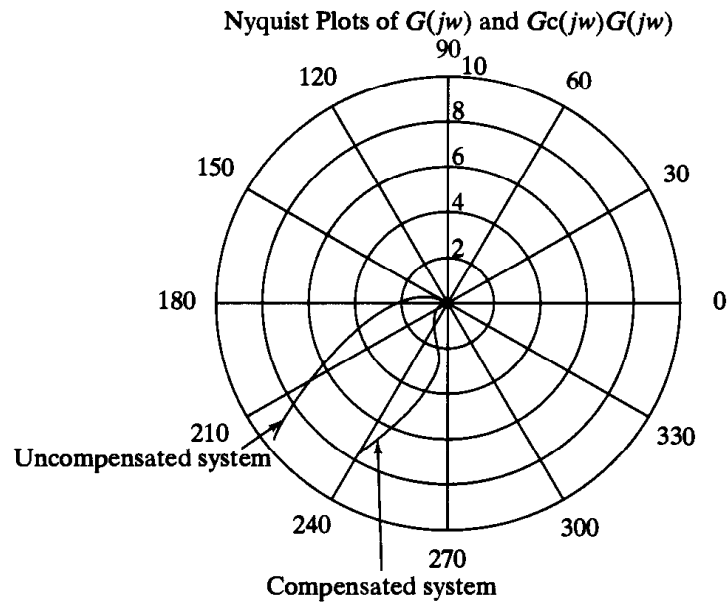
we have

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{G_c(s)G(s)}{1 + G_c(s)G(s)} \\ &= \frac{40(s + 0.4)(s + 0.2)}{(s + 4)(s + 0.02)s(s + 1)(s + 4) + 40(s + 0.4)(s + 0.2)} \end{aligned}$$



**Figure 9-52**

Bode diagram of the open-loop transfer function  $G_c(s)G(s)$  of the compensated system.



**Figure 9–53**  
Nyquist plots of  
 $G(j\omega)$  and  
 $G_c(j\omega)G(j\omega)$ .

MATLAB Program 9–15

```
%*****Nyquist plots*****

num0 = [0 0 0 40];
den0 = [1 5 4 0];
num1 = [0 0 0 40 24 3.2];
den1 = [1 9.02 24.18 16.48 0.32 0];
w = 0.8:0.02:10;
ww = 0.2:0.02:10;
[re0,im0,w] = nyquist(num0,den0,w);
z0 = re0 + i*im0;
r0 = abs(z0);
theta0 = angle(z0);
polar(theta0,r0)
hold on
[re1,im1,ww] = nyquist(num1,den1,ww);
z1 = re1 + i*im1;
r1 = abs(z1);
theta1 = angle(z1);
polar(theta1,r1)
text(-8, -12.7,'Compensated system')
text(-18.8, -7,'Uncompensated system')
title('Nyquist Plots of  $G(j\omega)$  and  $G_c(j\omega)G(j\omega)$ ')
```

To determine the denominator polynomial with MATLAB, we may proceed as follows:  
Define

$$a(s) = (s + 4)(s + 0.02) = s^2 + 4.02s + 0.08$$

$$b(s) = s(s + 1)(s + 4) = s^3 + 5s^2 + 4s$$

$$c(s) = 40(s + 0.4)(s + 0.2) = 40s^2 + 24s + 3.2$$

Then we have

$$\begin{aligned} a &= [1 \quad 4.02 \quad 0.08] \\ b &= [1 \quad 5 \quad 4 \quad 0] \\ c &= [40 \quad 24 \quad 3.2] \end{aligned}$$

Using the following MATLAB program, we obtain the denominator polynomial.

```
a = [1  4.02  0.08];
b = [1  5  4  0];
c = [40  24  3.2];
p = [conv(a,b)] + [0  0  0  c]
p =
    1.0000    9.0200   24.1800   56.4800   24.3200    3.2000
```

MATLAB Program 9–16 is used to obtain the unit-step response of the compensated system. The resulting unit-step response curve is shown in Figure 9–54. (Note that the uncompensated system is unstable.)

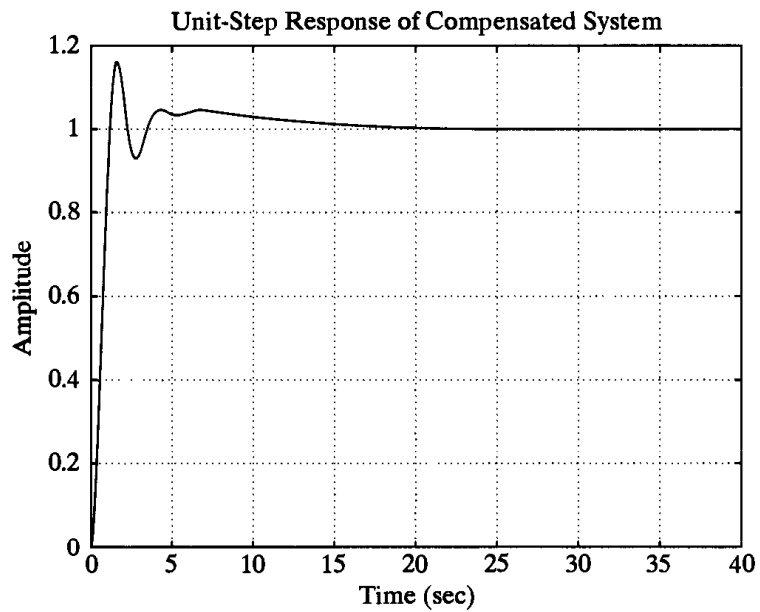
#### MATLAB Program 9–16

```
%*****Unit-step response*****

num = [0  0  0  40  24  3.2];
den = [1  9.02  24.18  56.48  24.32  3.2];
t = 0:0.2:40;
step(num,den,t)
grid
title('Unit-Step Response of Compensated System')
```

Unit-ramp response: The unit-ramp response of this system may be obtained by entering MATLAB Program 9–17 into the computer. Here we converted the unit-ramp response of  $G_c G / (1 + G_c G)$  into the unit-step response of  $G_c G / [s(1 + G_c G)]$ . The unit-ramp response curve obtained using this program is shown in Figure 9–55.

**Figure 9-54**  
Unit-step response  
curve of the compensated  
system.

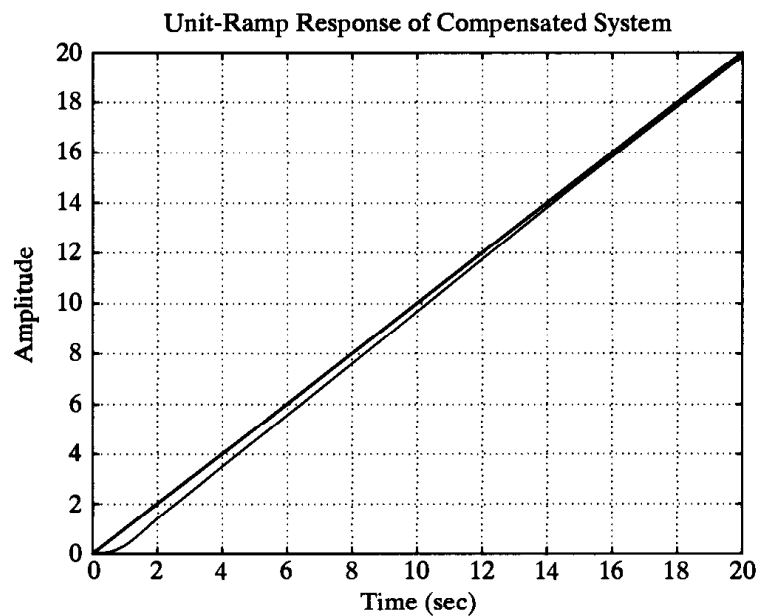


MATLAB Program 9-17

%\*\*\*\*\*Unit-ramp response\*\*\*\*\*

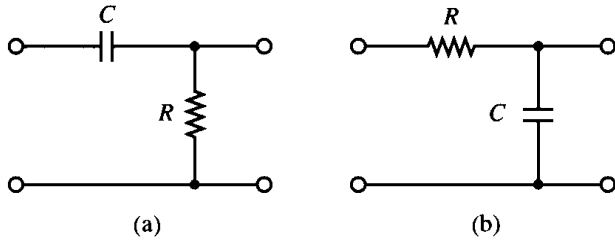
```
num = [0 0 0 0 40 24 3.2];
den = [1 9.02 24.18 56.48 24.32 3.2 0];
t = 0:0.05:20;
c = step(num,den,t);
plot(t,c,'-','t,t','.')
grid
title('Unit-Ramp Response of Compensated System')
xlabel('Time (sec)')
ylabel('Amplitude')
```

**Figure 9-55**  
Unit-ramp response  
of the compensated  
system.



## PROBLEMS

**B-9-1.** Draw Bode diagrams of the lead network and lag network shown in Figures 9-56 (a) and (b), respectively.



**Figure 9-56** (a) Lead network; (b) lag network.

**B-9-2.** Draw Bode diagrams of the PI controller given by

$$G_c(s) = 5 \left( 1 + \frac{1}{2s} \right)$$

and the PD controller given by

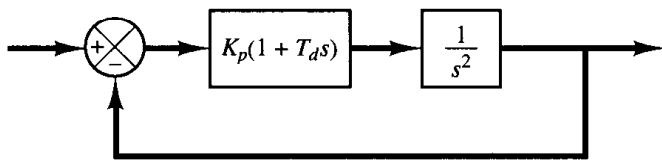
$$G_c(s) = 5(1 + 0.5s)$$

**B-9-3.** Consider a PID controller given by

$$G_c(s) = 30.3215 \frac{(s + 0.65)^2}{s}$$

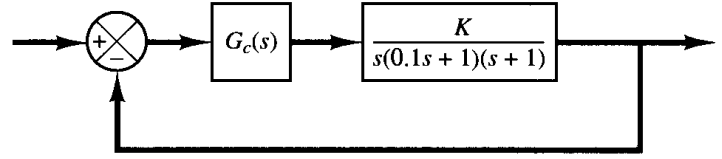
Draw a Bode diagram of the controller.

**B-9-4.** Figure 9-57 shows a block diagram of a space vehicle attitude control system. Determine the proportional gain constant  $K_p$  and derivative time  $T_d$  such that the bandwidth of the closed-loop system is 0.4 to 0.5 rad/sec. (Note that the closed-loop bandwidth is close to the gain crossover frequency.) The system must have an adequate phase margin. Plot both the open-loop and closed-loop frequency response curves on Bode diagrams.



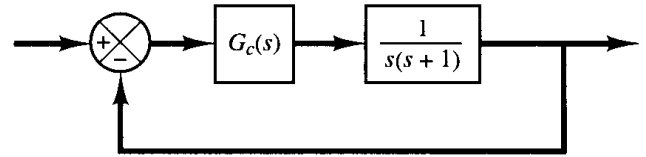
**Figure 9-57** Block diagram of space vehicle attitude control system.

**B-9-5.** Referring to the closed-loop system shown in Figure 9-58, design a lead compensator  $G_c(s)$  such that the phase margin is  $45^\circ$ , gain margin is not less than 8 dB, and the static velocity error constant  $K_v$  is  $4.0 \text{ sec}^{-1}$ . Plot unit-step and unit-ramp response curves of the compensated system with MATLAB.



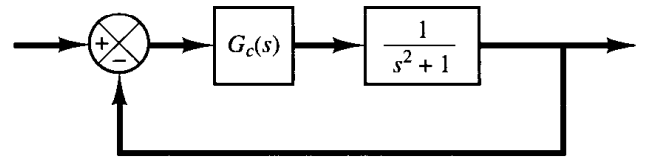
**Figure 9-58** Closed-loop system

**B-9-6.** Consider the system shown in Figure 9-59. Design a compensator such that the static velocity error constant  $K_v$  is  $50 \text{ sec}^{-1}$ , phase margin is  $50^\circ$ , and gain margin not less than 8 dB. Plot unit-step and unit-ramp response curves of the compensated and uncompensated systems with MATLAB.



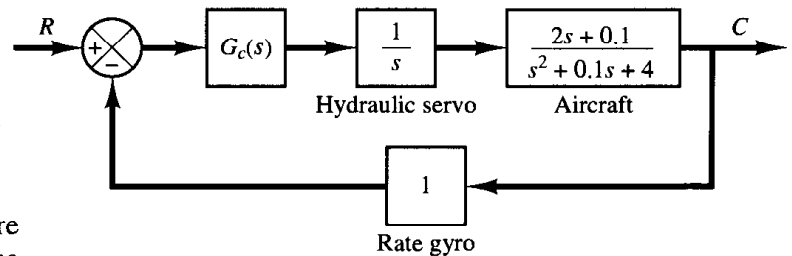
**Figure 9-59** Control system.

**B-9-7.** Consider the system shown in Figure 9-60. Design a compensator such that the static velocity error constant is  $4 \text{ sec}^{-1}$ , phase margin is  $50^\circ$ , and gain margin is 10 dB or more. Plot unit-step and unit-ramp response curves of the compensated system with MATLAB. Also, draw a Nyquist plot of the compensated system with MATLAB.



**Figure 9-60** Control system.

**B-9-8.** Consider the system shown in Figure 9-61. It is desired to design a compensator such that the static velocity



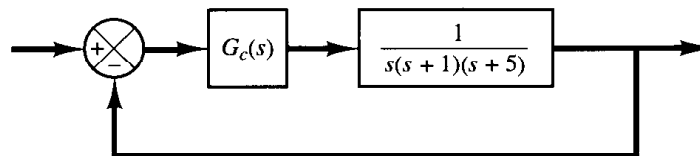
**Figure 9-61** Control system.



error constant is  $4 \text{ sec}^{-1}$ , phase margin is  $50^\circ$ , and gain margin is 8 dB or more. Plot the unit-step and unit-ramp response curves of the compensated system with MATLAB.

**B-9-9.** Consider the system shown in Figure 9-62. Design a lag-lead compensator such that the static velocity

error constant  $K_v$  is  $20 \text{ sec}^{-1}$ , phase margin is  $60^\circ$ , and gain margin is not less than 8 dB. Plot the unit-step and unit-ramp response curves of the compensated system with MATLAB.



**Figure 9-62** Control system.

# 10

## PID Controls and Introduction to Robust Control

### 10-1 INTRODUCTION

In previous chapters we occasionally discussed the basic PID control schemes. For example, in Chapter 5 we presented hydraulic, pneumatic, and electronic PID controllers. In Chapters 7 and 9 we designed control systems where PID controllers were involved.

In this chapter we first present tuning rules for the basic PID controllers and then discuss modified forms of PID control schemes, including PI-D control, I-PD control, and two-degrees-of-freedom PID control. Finally, we introduce the concept of robust design.

It is interesting to note that more than half of the industrial controllers in use today utilize PID or modified PID control schemes. Analog PID controllers are mostly hydraulic, pneumatic, electric, and electronic types or their combinations. Currently, many of these are transformed into digital forms through the use of microprocessors.

Because most PID controllers are adjusted on site, many different types of tuning rules have been proposed in the literature. Using these tuning rules, delicate and fine tuning of PID controllers can be made on site. Also, automatic tuning methods have been developed and some of the PID controllers may possess on-line automatic tuning capabilities. Modified forms of PID control, such as I-PD control and two-degrees-of-freedom PID control, are currently in use in industry. Many practical methods for bumpless switching (from manual operation to automatic operation) and gain scheduling are commercially available.

The usefulness of PID controls lies in their general applicability to most control systems. In the field of process control systems, it is a well-known fact that the basic and

modified PID control schemes have proved their usefulness in providing satisfactory control, although they may not provide optimal control in many given situations.

**Outline of the chapter.** Section 10–1 has presented introductory material for the chapter. Section 10–2 deals with tuning methods for the basic PID control, commonly known as Ziegler–Nichols tuning rules. Section 10–3 discusses modified PID control schemes, such as PI-D control and I-PD control. Section 10–4 introduces two-degrees-of-freedom PID control schemes. Section 10–5 introduces the concept of robust control using a two-degrees-of-freedom control system as an example.

## 10-2 TUNING RULES FOR PID CONTROLLERS

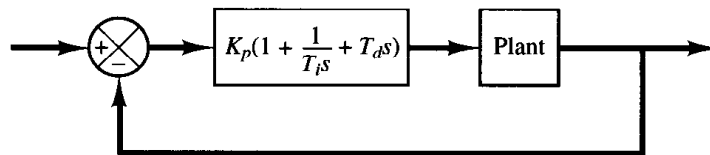
**PID control of plants.** Figure 10–1 shows a PID control of a plant. If a mathematical model of the plant can be derived, then it is possible to apply various design techniques for determining parameters of the controller that will meet the transient and steady-state specifications of the closed-loop system. However, if the plant is so complicated that its mathematical model cannot be easily obtained, then analytical approach to the design of a PID controller is not possible. Then we must resort to experimental approaches to the tuning of PID controllers.

The process of selecting the controller parameters to meet given performance specifications is known as controller tuning. Ziegler and Nichols suggested rules for tuning PID controllers (meaning to set values  $K_p$ ,  $T_i$ , and  $T_d$ ) based on experimental step responses or based on the value of  $K_p$  that results in marginal stability when only the proportional control action is used. Ziegler–Nichols rules, which are presented in the following, are very convenient when mathematical models of plants are not known. (These rules can, of course, be applied to the design of systems with known mathematical models.)

**Ziegler–Nichols rules for tuning PID controllers.** Ziegler and Nichols proposed rules for determining values of the proportional gain  $K_p$ , integral time  $T_i$ , and derivative time  $T_d$  based on the transient response characteristics of a given plant. Such determination of the parameters of PID controllers or tuning of PID controllers can be made by engineers on site by experiments on the plant. (Numerous tuning rules for PID controllers have been proposed since the Ziegler–Nichols proposal. They are available in the literature. Here, however, we introduce only the Ziegler–Nichols tuning rules.)

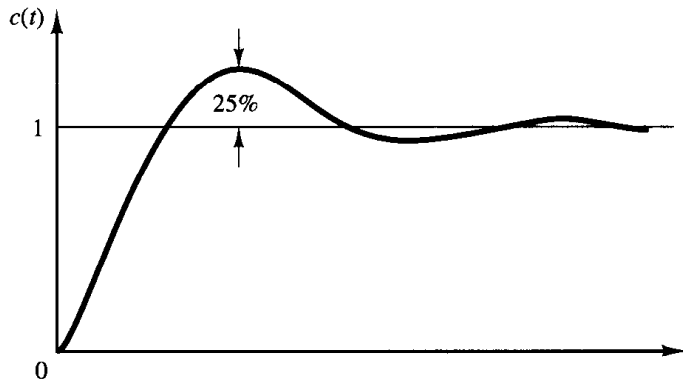
There are two methods called Ziegler–Nichols tuning rules. In both methods, they aimed at obtaining 25% maximum overshoot in step response (see Figure 10–2).

**First method.** In the first method, we obtain experimentally the response of the plant to a unit-step input, as shown in Figure 10–3. If the plant involves neither inte-

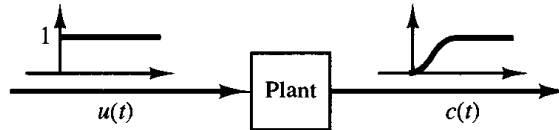


**Figure 10–1**  
PID control of a  
plant.

**Figure 10–2**  
Unit-step response curve showing 25% maximum overshoot.



**Figure 10–3**  
Unit-step response of a plant.



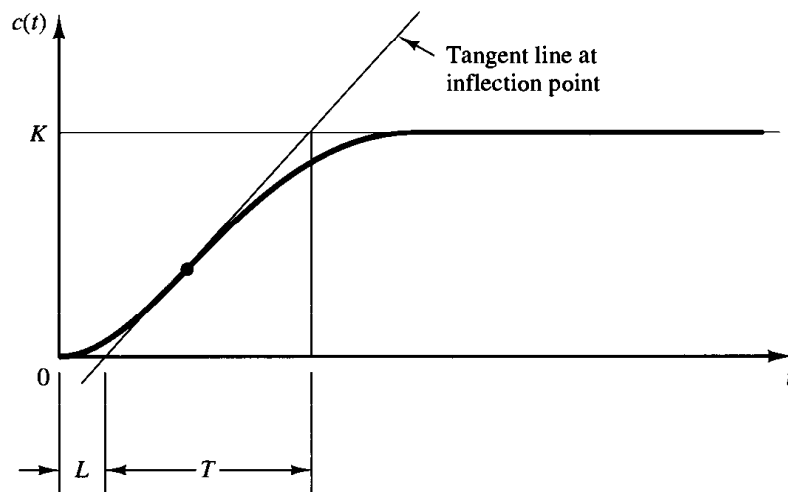
grator(s) nor dominant complex-conjugate poles, then such a unit-step response curve may look like an S-shaped curve, as shown in Figure 10–4. (If the response does not exhibit an S-shaped curve, this method does not apply.) Such step-response curves may be generated experimentally or from a dynamic simulation of the plant.

The S-shaped curve may be characterized by two constants, delay time  $L$  and time constant  $T$ . The delay time and time constant are determined by drawing a tangent line at the inflection point of the S-shaped curve and determining the intersections of the tangent line with the time axis and line  $c(t) = K$ , as shown in Figure 10–4. The transfer function  $C(s)/U(s)$  may then be approximated by a first-order system with a transport lag as follows:

$$\frac{C(s)}{U(s)} = \frac{Ke^{-Ls}}{Ts + 1}$$

Ziegler and Nichols suggested to set the values of  $K_p$ ,  $T_i$ , and  $T_d$  according to the formula shown in Table 10–1.

Notice that the PID controller tuned by the first method of Ziegler–Nichols rules gives



**Figure 10–4**  
S-shaped response curve.

**Table 10–1** Ziegler–Nichols Tuning Rule Based on Step Response of Plant (First Method)

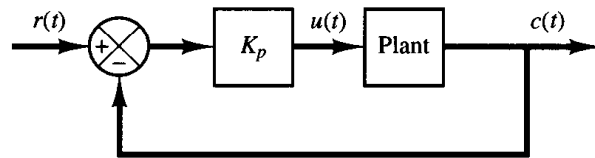
| Type of Controller | $K_p$             | $T_i$           | $T_d$  |
|--------------------|-------------------|-----------------|--------|
| P                  | $\frac{T}{L}$     | $\infty$        | 0      |
| PI                 | $0.9 \frac{T}{L}$ | $\frac{L}{0.3}$ | 0      |
| PID                | $1.2 \frac{T}{L}$ | $2L$            | $0.5L$ |

$$\begin{aligned}
 G_c(s) &= K_p \left( 1 + \frac{1}{T_i s} + T_d s \right) \\
 &= 1.2 \frac{T}{L} \left( 1 + \frac{1}{2Ls} + 0.5Ls \right) \\
 &= 0.6T \frac{\left( s + \frac{1}{L} \right)^2}{s}
 \end{aligned}$$

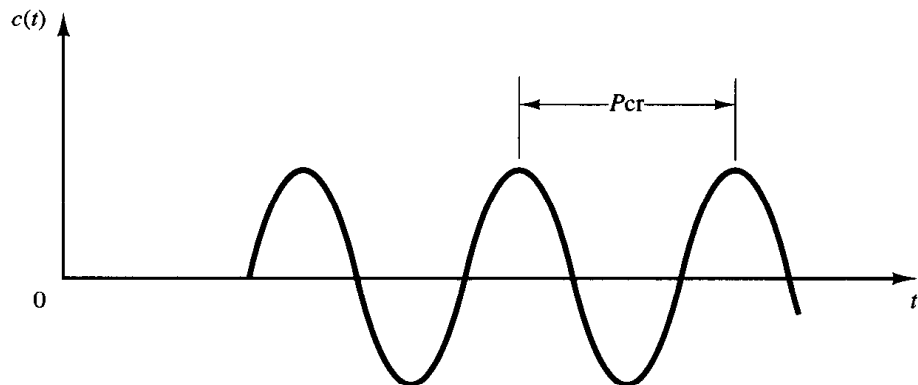
Thus, the PID controller has a pole at the origin and double zeros at  $s = -1/L$ .

**Second method.** In the second method, we first set  $T_i = \infty$  and  $T_d = 0$ . Using the proportional control action only (see Figure 10–5), increase  $K_p$  from 0 to a critical value  $K_{cr}$  where the output first exhibits sustained oscillations. (If the output does not exhibit sustained oscillations for whatever value  $K_p$  may take, then this method does not apply.) Thus, the critical gain  $K_{cr}$  and the corresponding period  $P_{cr}$  are experimentally determined (see Figure 10–6). Ziegler and Nichols suggested that we set the values of the parameters  $K_p$ ,  $T_i$ , and  $T_d$  according to the formula shown in Table 10–2.

**Figure 10–5**  
Closed-loop system with a proportional controller.



**Figure 10–6**  
Sustained oscillation with period  $P_{cr}$ .



**Table 10–2** Ziegler–Nichols Tuning Rule Based on Critical Gain  $K_{cr}$  and Critical Period  $P_{cr}$  (Second Method)

| Type of Controller | $K_p$        | $T_i$                 | $T_d$         |
|--------------------|--------------|-----------------------|---------------|
| P                  | $0.5K_{cr}$  | $\infty$              | 0             |
| PI                 | $0.45K_{cr}$ | $\frac{1}{1.2}P_{cr}$ | 0             |
| PID                | $0.6K_{cr}$  | $0.5P_{cr}$           | $0.125P_{cr}$ |

Notice that the PID controller tuned by the second method of Ziegler–Nichols rules gives

$$\begin{aligned}
 G_c(s) &= K_p \left( 1 + \frac{1}{T_i s} + T_d s \right) \\
 &= 0.6K_{cr} \left( 1 + \frac{1}{0.5P_{cr}s} + 0.125P_{cr}s \right) \\
 &= 0.075K_{cr}P_{cr} \frac{\left( s + \frac{4}{P_{cr}} \right)^2}{s}
 \end{aligned}$$

Thus, the PID controller has a pole at the origin and double zeros at  $s = -4/P_{cr}$ .

**Comments.** Ziegler–Nichols tuning rules (and other tuning rules presented in the literature) have been widely used to tune PID controllers in process control systems where the plant dynamics are not precisely known. Over many years, such tuning rules proved to be very useful. Ziegler–Nichols tuning rules can, of course, be applied to plants whose dynamics are known. (If plant dynamics are known, many analytical and graphical approaches to the design of PID controllers are available, in addition to Ziegler–Nichols tuning rules.)

If the transfer function of the plant is known, a unit-step response may be calculated or the critical gain  $K_{cr}$  and critical period  $P_{cr}$  may be calculated. Then, using those calculated values, it is possible to determine the parameters  $K_p$ ,  $T_i$ , and  $T_d$  from Table 10–1 or 10–2. However, the real usefulness of Ziegler–Nichols tuning rules (and other tuning rules) becomes apparent when the plant dynamics are not known so that no analytical or graphical approaches to the design of controllers are available.

Generally, for plants with complicated dynamics but no integrators, Ziegler–Nichols tuning rules can be applied. However, if the plant has an integrator, these rules may not be applied in some cases. To illustrate such a case where Ziegler–Nichols rules do not apply, consider the following case: Suppose that a unity-feedback control system has a plant whose transfer function is

$$G(s) = \frac{(s + 2)(s + 3)}{s(s + 1)(s + 5)}$$

Because of the presence of an integrator, the first method does not apply. Referring to

Figure 10–3, the step response of this plant will not have an S-shaped response curve; rather, the response increases with time. Also, if the second method is attempted (see Figure 10–5), the closed-loop system with a proportional controller will not exhibit sustained oscillations whatever value the gain  $K_p$  may take. This can be seen from the following analysis. Since the characteristic equation is

$$s(s + 1)(s + 5) + K_p(s + 2)(s + 3) = 0$$

or

$$s^3 + (6 + K_p)s^2 + (5 + 5K_p)s + 6K_p = 0$$

the Routh array becomes

|       |                                       |            |
|-------|---------------------------------------|------------|
| $s^3$ | 1                                     | $5 + 5K_p$ |
| $s^2$ | $6 + K_p$                             | $6K_p$     |
| $s^1$ | $\frac{30 + 29K_p + 5K_p^2}{6 + K_p}$ | 0          |
| $s^0$ | $6K_p$                                |            |

The coefficients in the first column are positive for all values of positive  $K_p$ . Thus, in the present case the closed-loop system will not exhibit sustained oscillations and, therefore, the critical gain value  $K_{cr}$  does not exist. Hence, the second method does not apply.

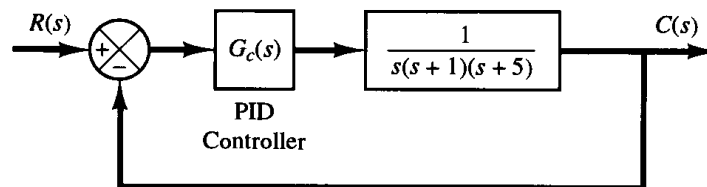
If the plant is such that Ziegler–Nichols rules can be applied, then the plant with a PID controller tuned by Ziegler–Nichols rules will exhibit approximately 10% ~ 60% maximum overshoot in step response. On the average (experimented on many different plants), the maximum overshoot is approximately 25%. (This is quite understandable because the values suggested in Tables 10–1 and 10–2 are based on the average.) In a given case, if the maximum overshoot is excessive, it is always possible (experimentally or otherwise) to make fine tuning so that the closed-loop system will exhibit satisfactory transient responses. In fact, Ziegler–Nichols tuning rules give an educated guess for the parameter values and provide a starting point for fine tuning.

#### EXAMPLE 10–1

Consider the control system shown in Figure 10–7 in which a PID controller is used to control the system. The PID controller has the transfer function

$$G_c(s) = K_p \left( 1 + \frac{1}{T_i s} + T_d s \right)$$

Although many analytical methods are available for the design of a PID controller for the present system, let us apply a Ziegler–Nichols tuning rule for the determination of the values of parameters  $K_p$ ,  $T_i$ , and  $T_d$ . Then obtain a unit-step response curve and check to see if the designed system exhibits approximately 25% maximum overshoot. If the maximum overshoot is excessive (40% or more), make a fine tuning and reduce the amount of the maximum overshoot to approximately 25%.



**Figure 10–7**  
PID-controlled  
system.

Since the plant has an integrator, we use the second method of Ziegler–Nichols tuning rules. By setting  $T_i = \infty$  and  $T_d = 0$ , we obtain the closed-loop transfer function as follows:

$$\frac{C(s)}{R(s)} = \frac{K_p}{s(s+1)(s+5) + K_p}$$

The value of  $K_p$  that makes the system marginally stable so that sustained oscillation occurs can be obtained by use of Routh's stability criterion. Since the characteristic equation for the closed-loop system is

$$s^3 + 6s^2 + 5s + K_p = 0$$

the Routh array becomes as follows:

$$\begin{array}{ccc} s^3 & 1 & 5 \\ s^2 & 6 & K_p \\ s^1 & \frac{30 - K_p}{6} & \\ s^0 & K_p & \end{array}$$

Examining the coefficients of the first column of the Routh table, we find that sustained oscillation will occur if  $K_p = 30$ . Thus, the critical gain  $K_{cr}$  is

$$K_{cr} = 30$$

With gain  $K_p$  set equal to  $K_{cr}$  ( $= 30$ ), the characteristic equation becomes

$$s^3 + 6s^2 + 5s + 30 = 0$$

To find the frequency of the sustained oscillation, we substitute  $s = j\omega$  into this characteristic equation as follows:

$$(j\omega)^3 + 6(j\omega)^2 + 5(j\omega) + 30 = 0$$

or

$$6(5 - \omega^2) + j\omega(5 - \omega^2) = 0$$

from which we find the frequency of the sustained oscillation to be  $\omega^2 = 5$  or  $\omega = \sqrt{5}$ . Hence, the period of sustained oscillation is

$$P_{cr} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{5}} = 2.8099$$

Referring to Table 10–2, we determine  $K_p$ ,  $T_i$ , and  $T_d$  as follows:

$$K_p = 0.6K_{cr} = 18$$

$$T_i = 0.5P_{cr} = 1.405$$

$$T_d = 0.125P_{cr} = 0.35124$$

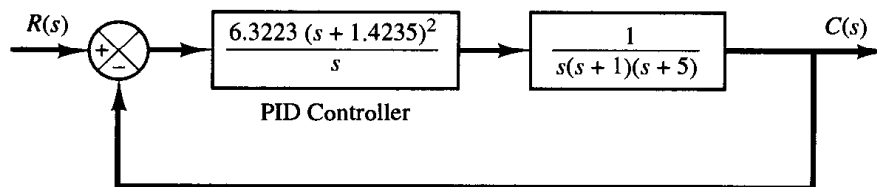
The transfer function of the PID controller is thus

$$\begin{aligned} G_c(s) &= K_p \left( 1 + \frac{1}{T_i s} + T_d s \right) \\ &= 18 \left( 1 + \frac{1}{1.405s} + 0.35124s \right) \\ &= \frac{6.3223(s + 1.4235)^2}{s} \end{aligned}$$



**Figure 10–8**

Block diagram of the system with PID controller designed by use of Ziegler–Nichols tuning rule (second method).



The PID controller has a pole at the origin and double zero at  $s = -1.4235$ . A block diagram of the control system with the designed PID controller is shown in Figure 10–8.

Next, let us examine the unit-step response of the system. The closed-loop transfer function  $C(s)/R(s)$  is given by

$$\frac{C(s)}{R(s)} = \frac{6.3223s^2 + 18s + 12.811}{s^4 + 6s^3 + 11.3223s^2 + 18s + 12.811}$$

The unit-step response of this system can be obtained easily with MATLAB. See MATLAB Program 10–1. The resulting unit-step response curve is shown in Figure 10–9. The maximum overshoot in the unit-step response is approximately 62%. The amount of maximum overshoot is excessive. It can be reduced by fine tuning the controller parameters. Such fine tuning can be made on the computer. We find that by keeping  $K_p = 18$  and by moving the double zero of the PID controller to  $s = -0.65$ , that is, using the PID controller

$$G_c(s) = 18 \left( 1 + \frac{1}{3.077s} + 0.7692s \right) = 13.846 \frac{(s + 0.65)^2}{s} \quad (10-1)$$

the maximum overshoot in the unit-step response can be reduced to approximately 18% (see Figure 10–10). If the proportional gain  $K_p$  is increased to 39.42, without changing the location of the double zero ( $s = -0.65$ ), that is, using the PID controller

$$G_c(s) = 39.42 \left( 1 + \frac{1}{3.077s} + 0.7692s \right) = 30.322 \frac{(s + 0.65)^2}{s} \quad (10-2)$$

then the speed of response is increased, but the maximum overshoot is also increased to approximately 28%, as shown in Figure 10–11. Since the maximum overshoot in this case is fairly close to 25% and the response is faster than the system with  $G_c(s)$  given by Equation (10–1), we may consider  $G_c(s)$  as given by Equation (10–2) as acceptable. Then the tuned values of  $K_p$ ,  $T_i$ , and  $T_d$  become

$$K_p = 39.42, \quad T_i = 3.077, \quad T_d = 0.7692$$

It is interesting to observe that these values respectively are approximately twice the values suggested by the second method of the Ziegler–Nichols tuning rule. The important thing to note here is that the Ziegler–Nichols tuning rule has provided a starting point for fine tuning.

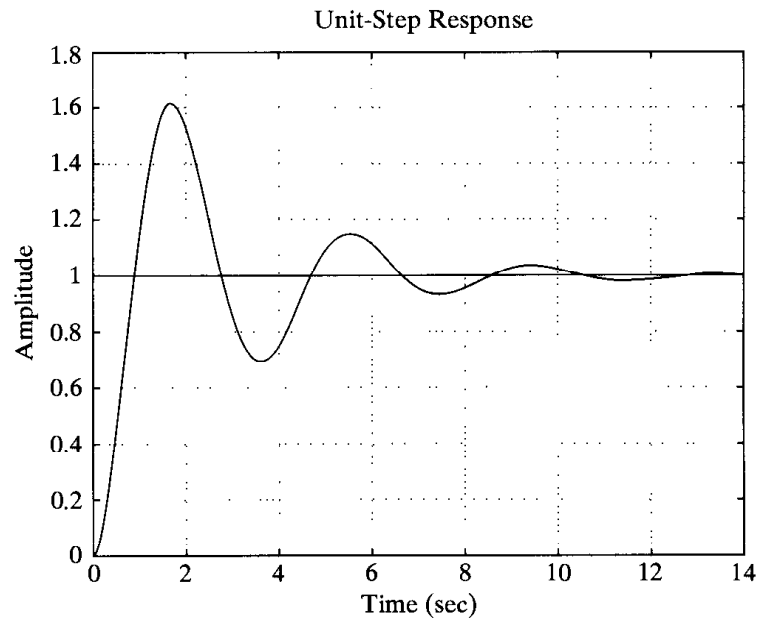
It is instructive to note that, for the case where the double zero is located at  $s = -1.4235$ , increasing the value of  $K_p$  increases the speed of response, but as far as the percentage maximum

#### MATLAB Program 10–1

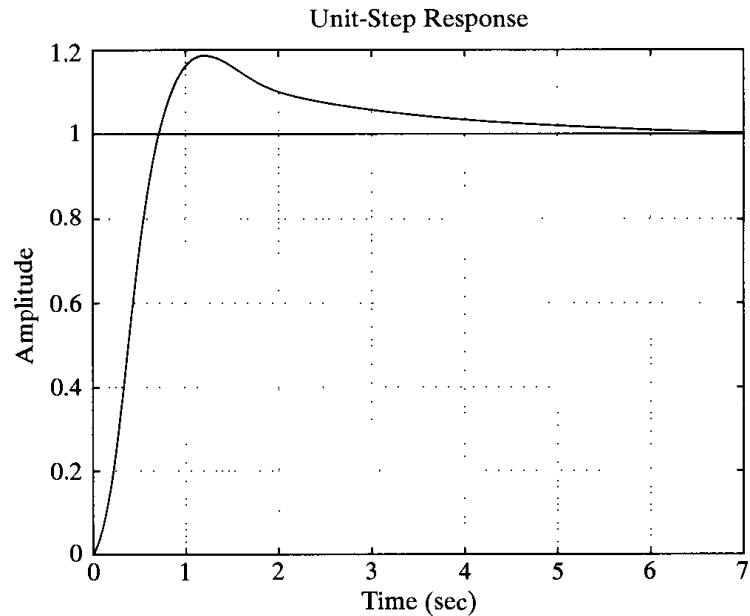
% ----- Unit-step response -----

```
num = [0 0 6.3223 18 12.811];
den = [1 6 11.3223 18 12.811];
step(num,den)
grid
title('Unit-Step Response')
```

**Figure 10-9**  
Unit-step response  
curve of PID-  
controlled system  
designed by use of  
Ziegler–Nichols  
tuning rule (second  
method).



**Figure 10-10**  
Unit-step response  
of the system  
shown in Figure  
10-7 with PID con-  
troller having para-  
meters  $K_p = 18$ ,  
 $T_i = 3.077$ , and  
 $T_d = 0.7692$ .



overshoot is concerned, varying gain  $K_p$  has very little effect. The reason for this may be seen from the root-locus analysis. Figure 10-12 shows the root-locus diagram for the system designed by use of the second method of Ziegler–Nichols tuning rules. Since the dominant branches of root loci are along the  $\zeta = 0.3$  lines for a considerable range of  $K$ , varying the value of  $K$  (from 6 to 30) will not change the damping ratio of the dominant closed-loop poles very much. However, varying the location of the double zero has a significant effect on the maximum overshoot, because the damping ratio of the dominant closed-loop poles can be changed significantly. This can also be seen from the root-locus analysis. Figure 10-13 shows the root-locus diagram for the system where the PID controller has the double zero at  $s = -0.65$ . Notice the change of the root-locus configuration. This change in the configuration makes it possible to change the damping ratio of the dominant closed-loop poles.